Methods to Find Limits

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Method 1: Substitution

- If the function is continuous at the limit, then sub in the limit

Example:

$\lim_{x \to 2} x^2 = 2^2 = 4$

- Doesn't work if you get an undefined result (e.g $\frac{0}{0}$ or $\frac{\infty}{\infty}$)

Method 2: Simplify/Factorise then Substitute

- Cancel out problematic terms that prevent method 1 from working

Example:

$\lim_{x \to 1} \frac{x^2+x-2}{x^2+2x-3} \rightarrow$ factorise to $\frac{(x-1)(x+2)}{(x-1)(x+3)}$

and cancel out $(x-1)$ to get $\frac{x+2}{x+3}$

— Method 1 —

$\lim_{x \to 1} \frac{x+2}{x+3} = \boxed{\frac{3}{4}}$

Method 3: The Pinching Theorem (Squeeze Theorem)

- Used when a function oscillates wildly, but is "trapped" by two other functions going to the same limit

Logic:

- when trying to find the limit f
- If $g(x) \leq f(x) \leq h(x)$, AND both the limits of g and h are L
- Then the limit of f is also L

Example:

$\lim_{x \to 0} x\sin\left(\frac{1}{x}\right)$

Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$, $-|x| \leq x\sin\left(\frac{1}{x}\right) \leq |x|$

As both $-|x|$ and $|x|$ go to 0 at $\lim_{x \to 0}$,

$x\sin\left(\frac{1}{x}\right)$ also goes to 0 at $\lim_{x \to 0}$

$$\boxed{\therefore \lim_{x \to 0} x\sin\left(\frac{1}{x}\right) = 0}$$

Method 4: Standard Limits (Memorise)

- Common Limits to memorise
- Trigonometric:
  - $\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \xrightarrow{\text{implies}} \lim_{x \to 0} \frac{\tan(x)}{x} = 1$
- Exponential:
  - $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$
- Logarithmic:
  - $\lim_{x \to \infty} \frac{1}{\ln(x)} = 0$

Method 5: Dividing by the Highest Power

- Used for Limits at infinity
- Divide every term by the highest power of x in the denominator

Example:

$$\lim_{x \to \infty} \frac{2x^2 - 7x}{3x^2 + 4} \xrightarrow{\text{Divide by } x^2} \frac{2 - \frac{7}{x}}{3 + \frac{4}{x^2}}$$

As $x \to \infty$, $\frac{2 - \frac{7}{x}}{3 + \frac{4}{x^2}} \to \boxed{\frac{2}{3}}$

- Note that this method doesn't work for polynomials (only fractions)

Method 6: Substituting Variables

- Substituting variables to make a standard limit recognisable

Example:

$$\lim_{x \to 0} \frac{\sin(3x)}{x}$$

let $y = 3x \to x = \frac{1}{3}y$

$$\frac{\sin(3x)}{x} = \frac{\sin(y)}{\frac{1}{3}y} = 3\frac{\sin(y)}{y} = 3(1) = \boxed{3}$$

Method 7: L'Hôpital's Rule

The Conditions:

- Limit produces a $\frac{0}{0}$ or $\frac{\infty}{\infty}$

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- The derivatives of the top and bottom **exist** and are **continuous**

Algorithm:

1) Check if Limit gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$
2) Differentiate top and bottom (seperately not Quotient Rule)
3) Find $\lim_{x \to a} \frac{g'(x)}{h'(x)}$
4) If the result is still $\frac{0}{0}$, you can apply the rule again

Example:

Question: Find $\lim_{x \to 0} \frac{\cos(x) - 1}{x^2}$

- Check: $\cos(0) - 1 = 0$, $0^2 = 0$ $\therefore$ we have $\frac{0}{0}$
- Apply L'Hôpital:
  - Differentiate top $(-\sin(x))$ and bottom $(2x)$
  - New Limit: $\lim_{x \to 0} \frac{-\sin(x)}{2x}$
- Check again: $-\sin(0) = 0$, $2(0) = 0$ $\therefore$ still $\frac{0}{0}$
- Apply L'Hôpital:
  - Differentiate top $(-\cos(x))$ and bottom (2)
- Final Limit:

$$\lim_{x \to 0} \frac{-\cos(x)}{2} = -\frac{1}{2}$$

Using Limits to Check for Continuity:

- A function $f(x)$ is continuous at $x = a$,

$$\boxed{\text{if } \lim_{x \to a} f(x) = f(a)}$$

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(and $f(a)$ exists)